

SEIZURE DETECTION IN EPILEPSY PATIENTS USING MACHINE LEARNING ALGORITHMS



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ABSTRACT

There are over 50 million people who currently suffer from epilepsy^[1]. Seizures, the primary feature of this disease, are defined broadly as abnormal electrical activity in the brain during a certain period of time. Electroencephalography (EEG) is a test that measures the electrical activity in the brain and is used in the field of seizure onset detection. The primary focus of this project is to develop a patient-specific machine learning model that can efficiently classify whether or not an event is a seizure using EEG data. At the end of the project, a model was created which had a seizure accuracy rate of 69%, a non-seizure accuracy rate of 98%, and a latency period of 0.48 seconds when tested on an hour of EEG data.

BACKGROUND

- EEG processing follows a general outline (Figure 1)
- Preprocessing aims to refine raw EEG data
- Feature extraction and selection highlights the imperceptible information from the data
- EEG data is filtered into frequency bands^[2]: delta, theta, alpha, beta, and gamma (Figure 2)
- The power level of these frequency bands change rapidly during a seizure
- A machine learning classification model can be created with this data

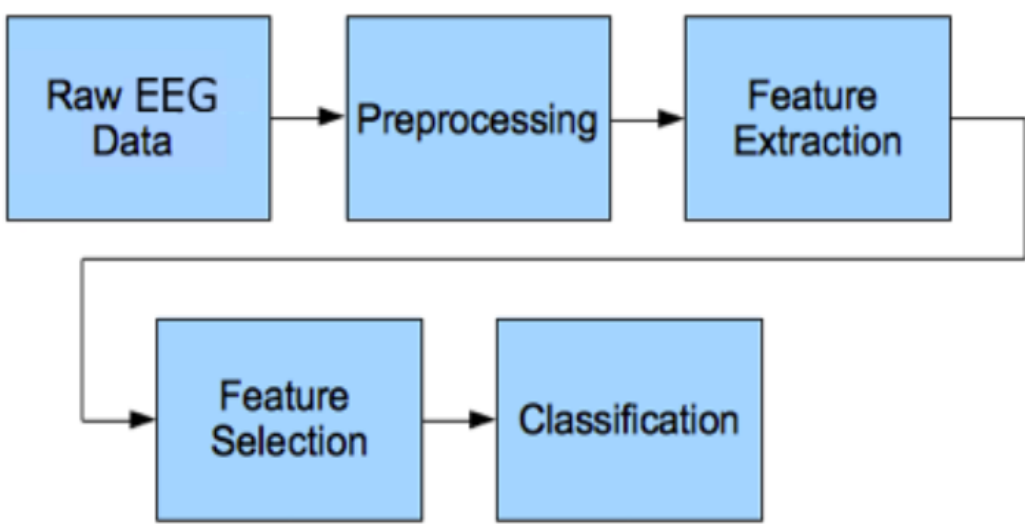


Figure 1: General outline of EEG data processing.

Waves	Frequency bands (Hz)	Behaviour Trait	Signal Waveform
Delta	0.3 – 4	Deep sleep	
Theta	4 – 8	Deep Meditation	
Alpha	8 – 13	Eyes closed, awake	
Beta	13 – 30	Eyes opened, thinking	
Gamma	30 and above	Unifying consciousness	

Figure 2: The different frequency bands.

EXPERIMENTAL METHODS

1. Filter data into different frequency bands and convert into power (watts) (Figure 3)
2. Use power from each frequency band as a feature to be fed to the classifier
3. Create 92 column feature vector from the 23 channels and label it with an additional column with 0 (non-seizure) and 1 (seizure) (Figure 4)
4. Train and test linear Support Vector Machine (SVM) to classify the points in feature vector (Figure 5)

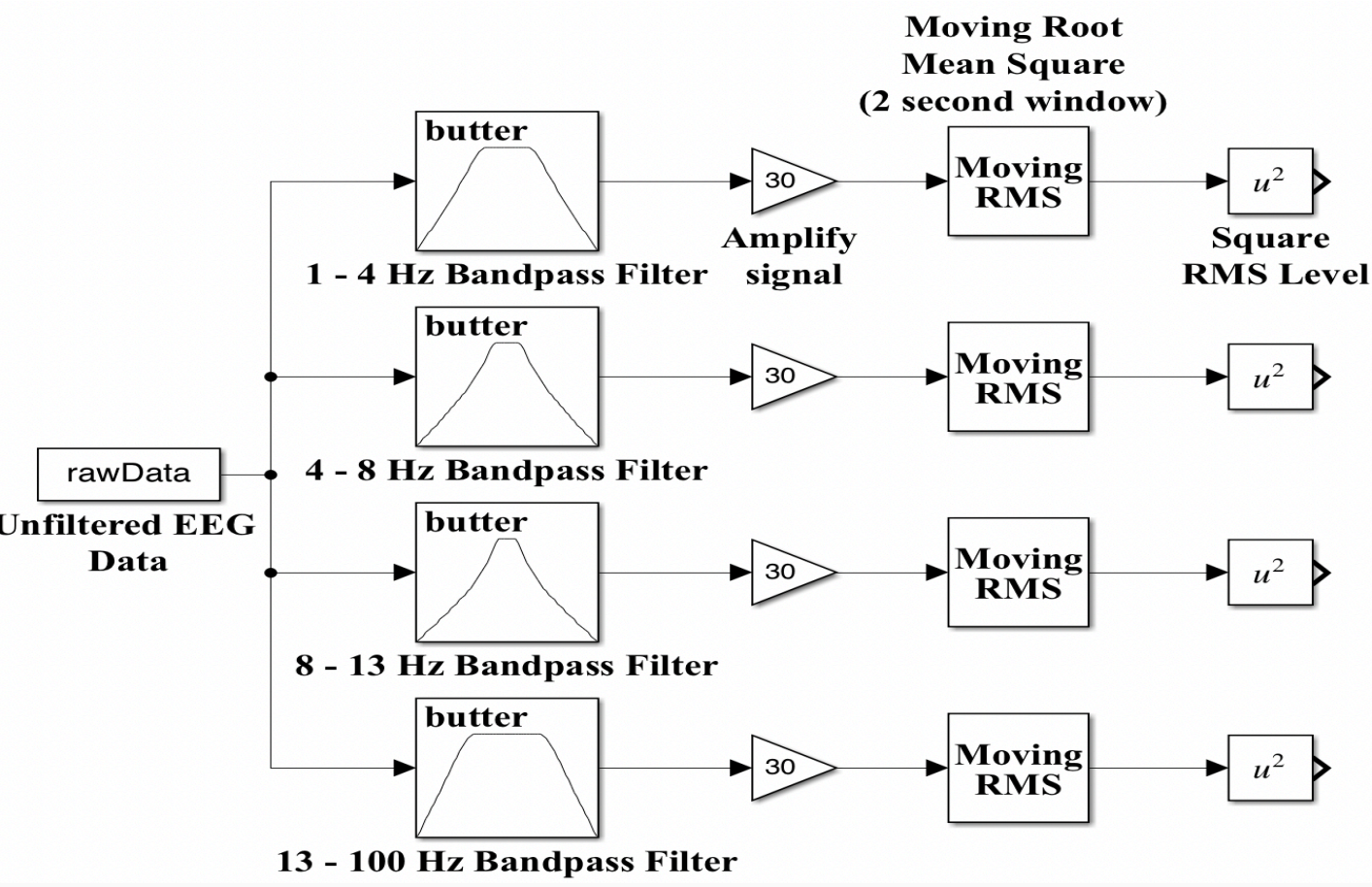


Figure 3 (left): Block diagram created in Simulink to calculate power of each frequency.

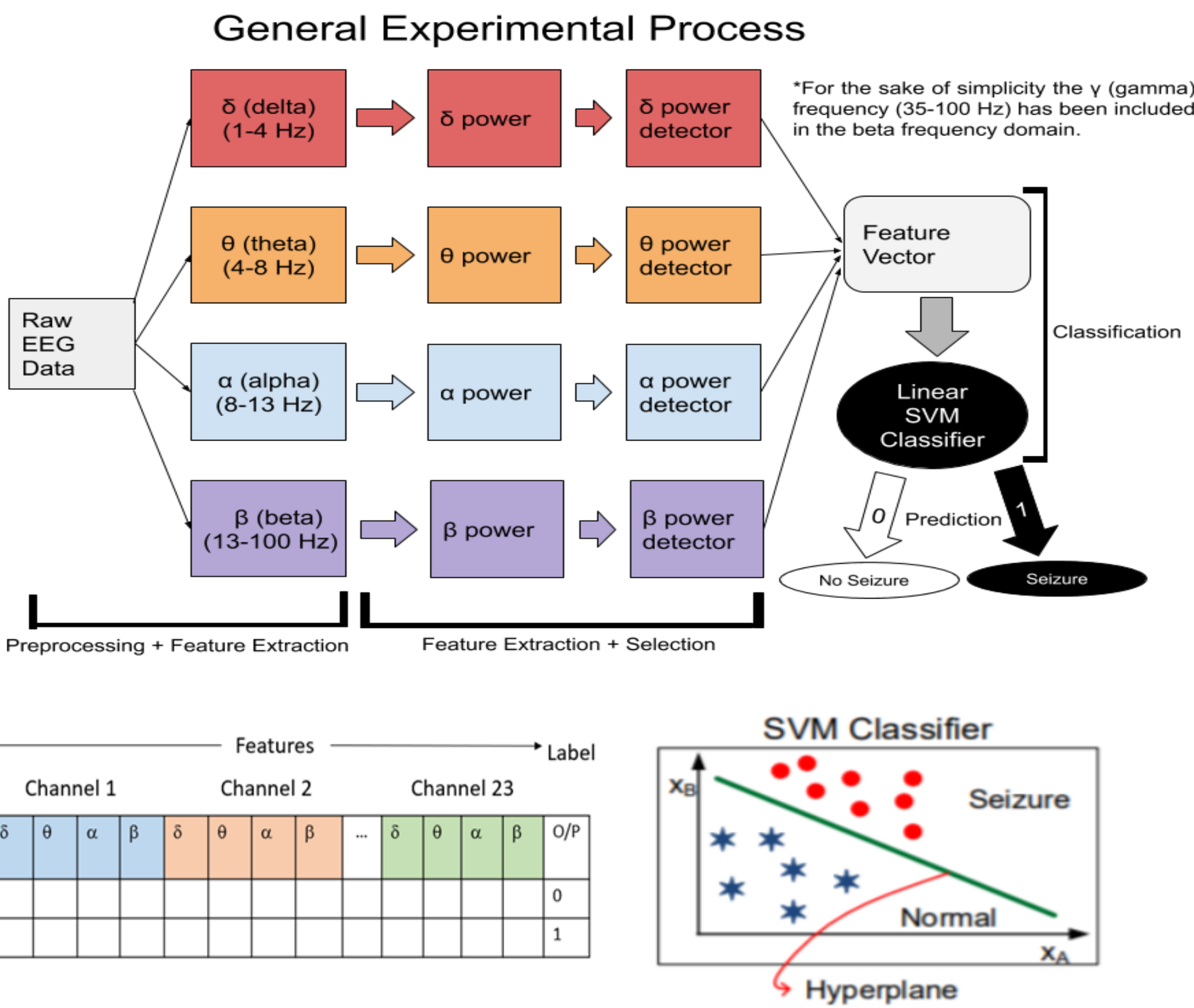


Figure 4: A general diagram of the final labeled 93-column feature vector.

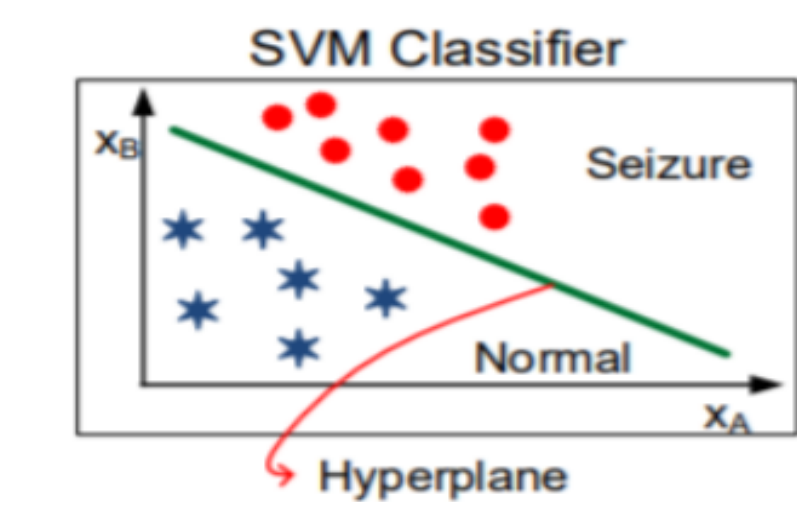


Figure 5: A hyperplane is produced by a SVM that classifies data points.

RESULTS

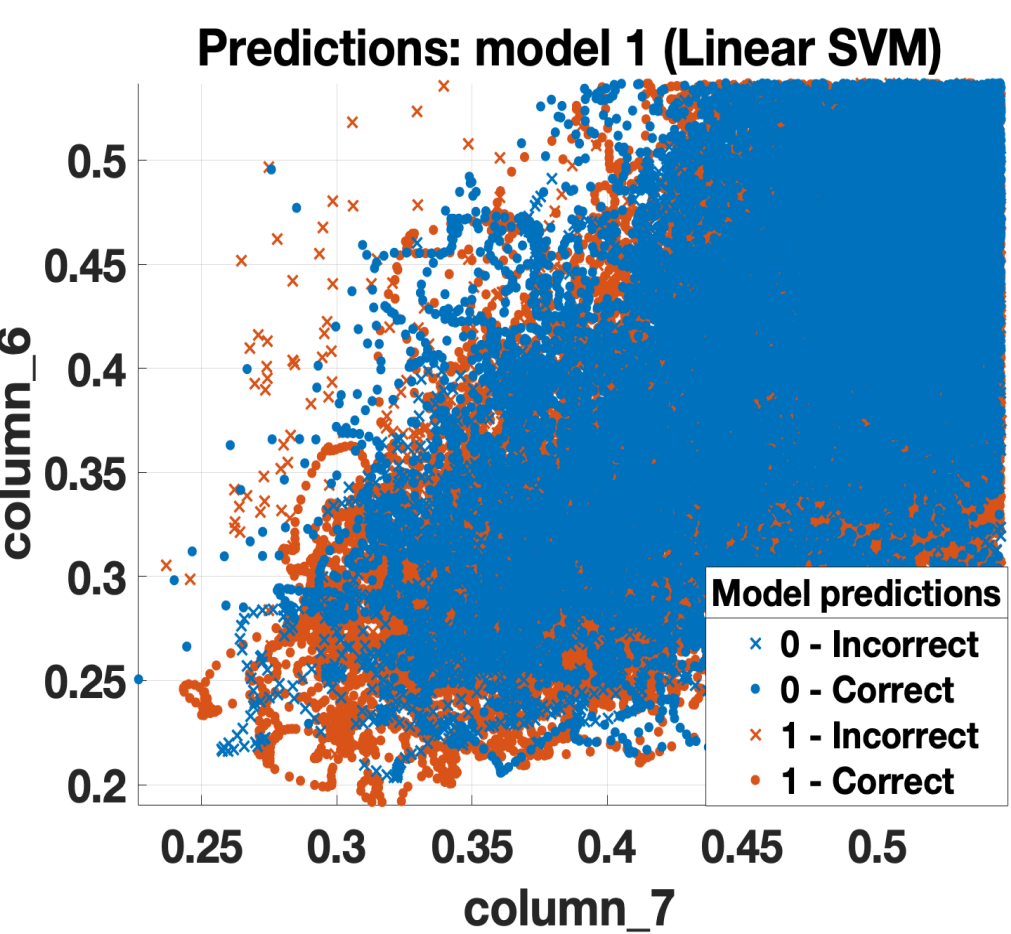


Figure 6 (above): A scatter plot of the predictions.

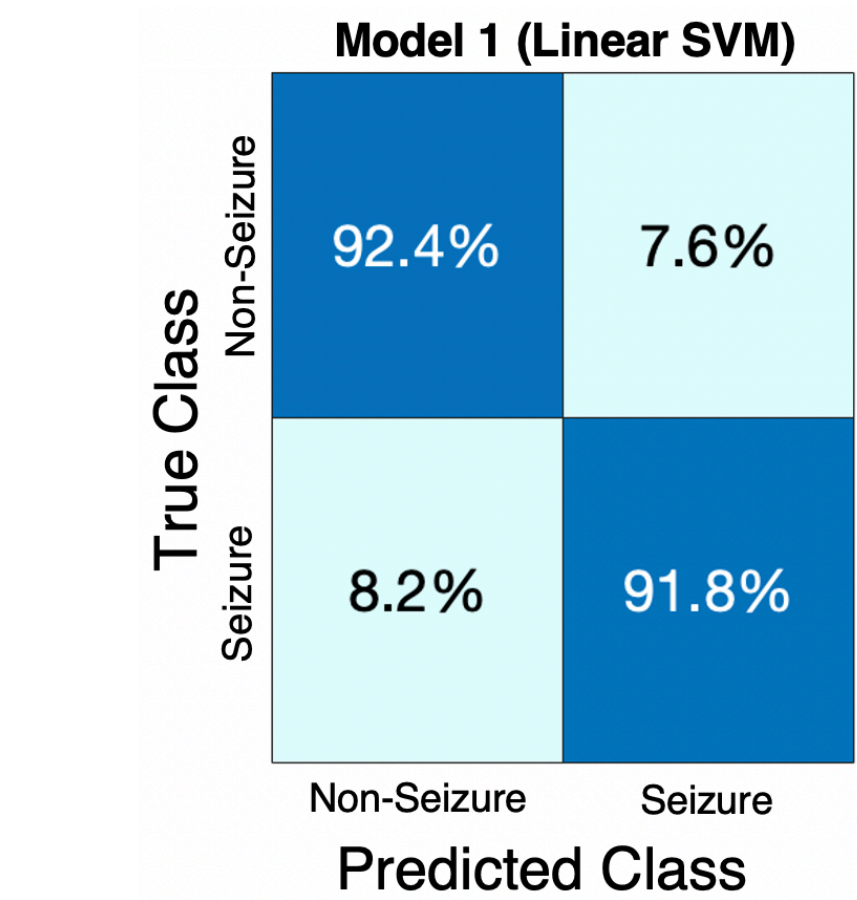


Figure 7 (above): Confusion matrix for training data set.

	Non-Seizure	Seizure	Non-Seizure and Seizure
Correct	97.87%	69.16%	97.54%
Incorrect	2.13%	30.84%	2.46%
Seizure Detection Latency	0.48 seconds		

Figure 8: The predictions of our SVM on an hour of EEG data after being trained with 41 hours of data.

CONCLUSION

- Improve classification accuracy by:
 - Training model with more data
 - Adding temporal data as a feature
- Implement model onto implantable onset seizure detection device
- Create model/classifier to predict rather than detect seizures

	This Work	[3]	[4]	[5]
Seizures Detected (%)	69	84	95	96
Channels	23	8	8	18

Figure 9: Feature extraction parameter comparison.

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